

IIDS67692 Computational Methods for Multimodal Data Analysis

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Part A: (word counts: 498)

Methods

Spatial Graph Convolutional Network (SpaGCN) is a graph-based deep learning framework for spatial transcriptomics analysis (Hu et al., 2021). It performs **early fusion** by combining three modalities — gene expression counts, spatial coordinates, and histology image pixel intensities — prior to any learning step. These signals are merged during graph construction: a weighted adjacency matrix encodes both Euclidean distance between tissue spots and morphological similarity derived from histology, with parameter s controlling the relative image contribution. Because all modalities are integrated before feature learning begins, the architecture constitutes an early fusion design, enabling each modality to directly shape the GCN latent representation. The fused graph is then processed by GCN layers and refined through iterative deep clustering (DEC) initialised by k-means.

The Visium DLPFC dataset (sample 151676) was loaded from filtered_feature_bc_matrix.h5, spatial barcodes, and the full-resolution histology image (151676_full_image.tif). Preprocessing involved extracting RGB pixel intensities at each spot's coordinate and constructing the spatial weight matrix. SpaGCN was applied under $s = 0, 1$, and 10 , followed by k-means clustering with 7 clusters matching the ground-truth labels (Layers 1–6 and White Matter). Clusters were visualised spatially and evaluated using ARI, FMI, AMI, and per-label F1-scores.

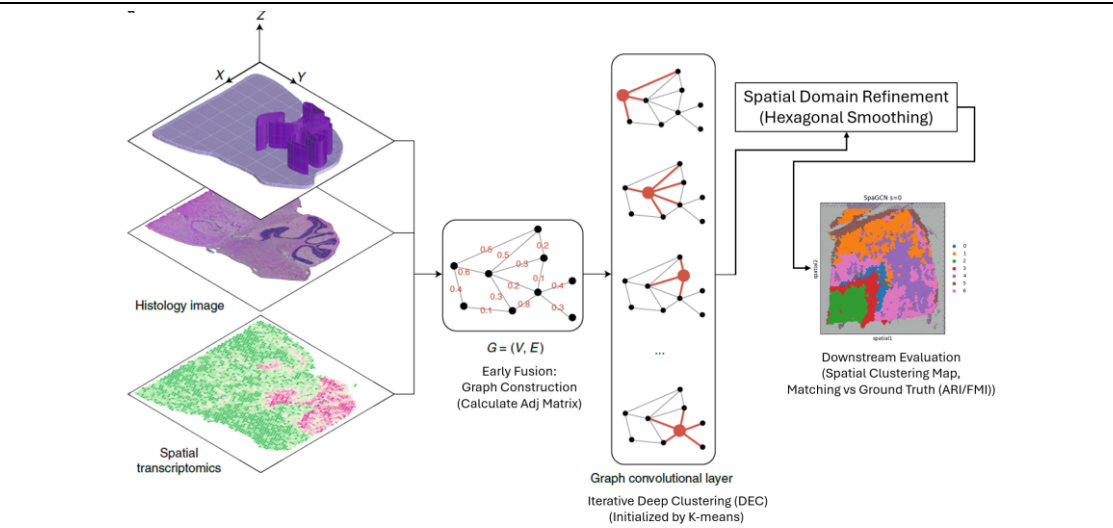


Figure 1. Overview of the SpaGCN Workflow.

Gene expression, histology, and spatial coordinate data undergo early fusion to build an adjacency graph. The integrated features are processed via graph convolutional layers and refined through hexagonal smoothing to delineate spatial domains. The final spatial clustering maps are utilised for downstream evaluation by benchmarking and matching against the annotated ground truth.

Results

Across all s values, Layer 1, White Matter (WM), and Layer 6 are consistently identified, as these structurally extreme regions are robustly anchored by both transcriptomic and morphological signals. Critical differences emerge in intermediate cortical layers (2–5).

At $s = 0$, transcriptomics-only analysis produces the highest global alignment (FMI = 0.3648, AMI = 0.3838), sharply separating WM (F1 = 0.84) and Layer 1 (F1 = 0.68). However, without morphological guidance, intermediate boundaries become noisy; Layer 4 is entirely missed (F1 = 0.00) because its transcriptomic signature is insufficiently distinct from adjacent layers. Convergence is rapid at 46 epochs.

At $s = 1$, moderate histological input acts as a spatial smoothing constraint, suppressing sequencing noise while preserving biological continuity. The model achieves the most anatomically faithful stratification of intermediate layers, though global metrics are moderate (ARI = 0.2105, FMI = 0.3457). WM (F1 = 0.76) and Layer 1 (F1 = 0.65) remain well recovered, but Layer 4 is again undetected (F1 = 0.00).

At $s = 10$, image features overwhelm transcriptomic signals, producing over-smoothed cluster boundaries and the lowest global metrics (FMI = 0.3425, AMI = 0.3564). Layer 1 (F1 = 0.43) and Layer 6 (F1 = 0.22) degrade as morphological entanglement absorbs adjacent spots. Critically, $s = 10$ is the only configuration recovering Layer 4 (F1 = 0.21), demonstrating that heavy histological weighting provides the texture-level contrast necessary to resolve this elusive stratum. GCN optimisation is hampered, requiring 121 epochs to converge.

Conclusion

No single s value is universally optimal. $s = 0$ maximises global accuracy for macro-structures; $s = 1$ provides the most anatomically faithful spatial continuity; $s = 10$ uniquely resolves subtle mid-layer boundaries at the cost of broader coherence. Parameter selection should be guided by clinical objectives: transcriptomics-only for structurally distinct boundaries, heavy histological weighting for granular mid-layer resolution.

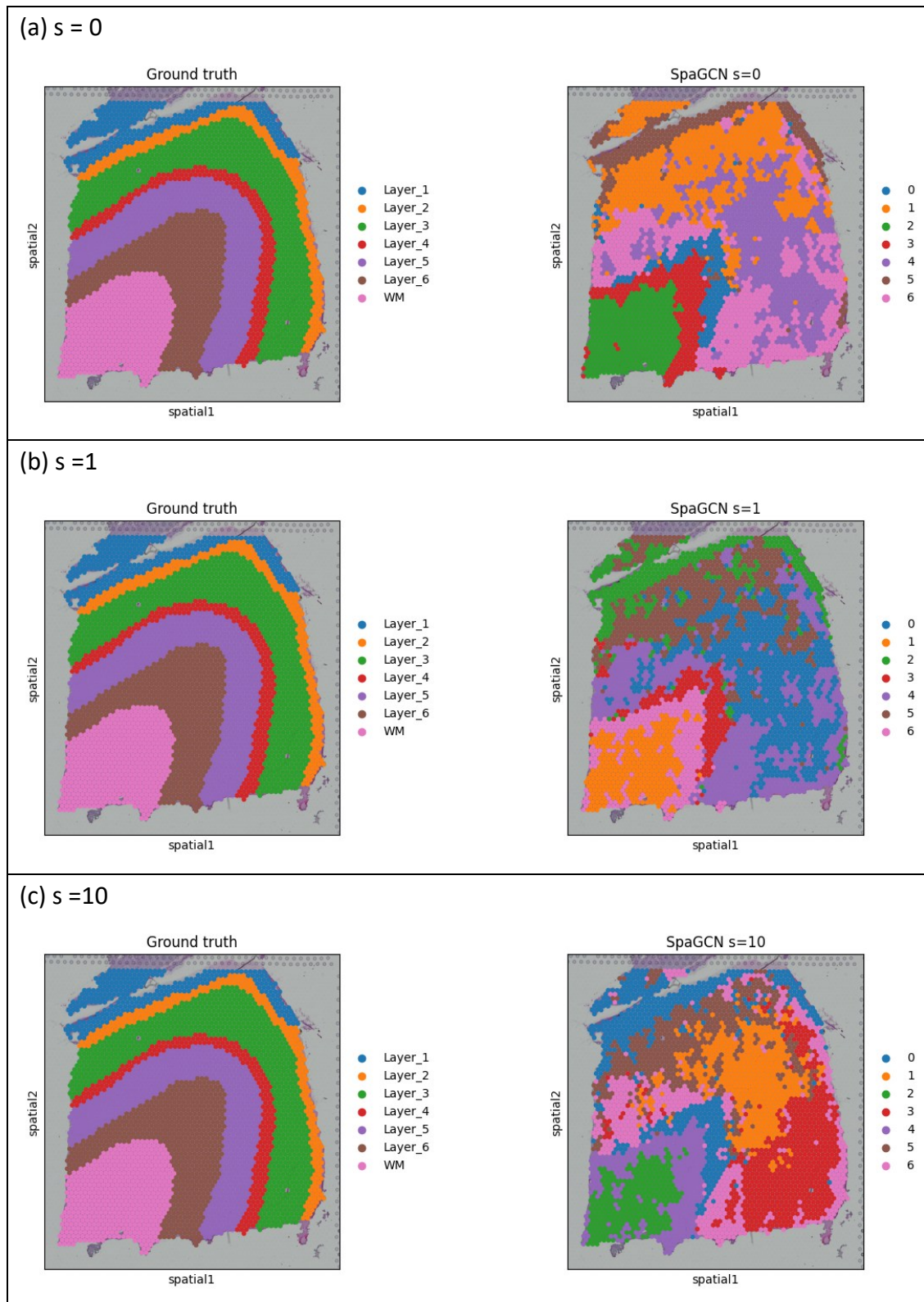


Figure 2. Spatial Clustering Maps Across Different Histology Contribution Settings.

The visual comparison illustrates how the histology weight parameter $s=0$, $s=1$, and $s=10$ affects spatial domain identification and cortical layer stratification, benchmarking the predicted spatial clusters against the manually annotated ground-truth map.

Table 1. Quantitative Evaluation and Per-Label Classification Performance Under Varying Histology Weights s .

The table presents global clustering metrics (FMI, ARI, AMI) and granular layer-by-layer classification reports (Precision, Recall, F1-score) to evaluate the impact of the image contribution parameter $s = 0, 1, 10$ on SpaGCN.

	$s = 0$			$s = 1$			$s = 10$			
	FMI	ARI	AMI	FMI	ARI	AMI	FMI	ARI	AMI	
	0.3648	0.2293	0.3838	0.3457	0.2105	0.3677	0.3425	0.2171	0.3564	
Layer	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Support
Layer_1	0.71	0.66	0.68	0.56	0.77	0.65	0.43	0.82	0.56	289
Layer_2	0.1	0.3	0.15	0.1	0.3	0.15	0.2	0.44	0.27	254
Layer_3	0.5	0.49	0.49	0.52	0.39	0.45	0.55	0.35	0.43	836
Layer_4	0	0	0	0	0	0	0.15	0.37	0.21	254
Layer_5	0.39	0.47	0.43	0.38	0.47	0.42	0.51	0.4	0.45	649
Layer_6	0.92	0.26	0.41	0.91	0.28	0.43	0.3	0.17	0.22	616
WM	0.99	0.72	0.84	0.98	0.62	0.76	1	0.56	0.71	533

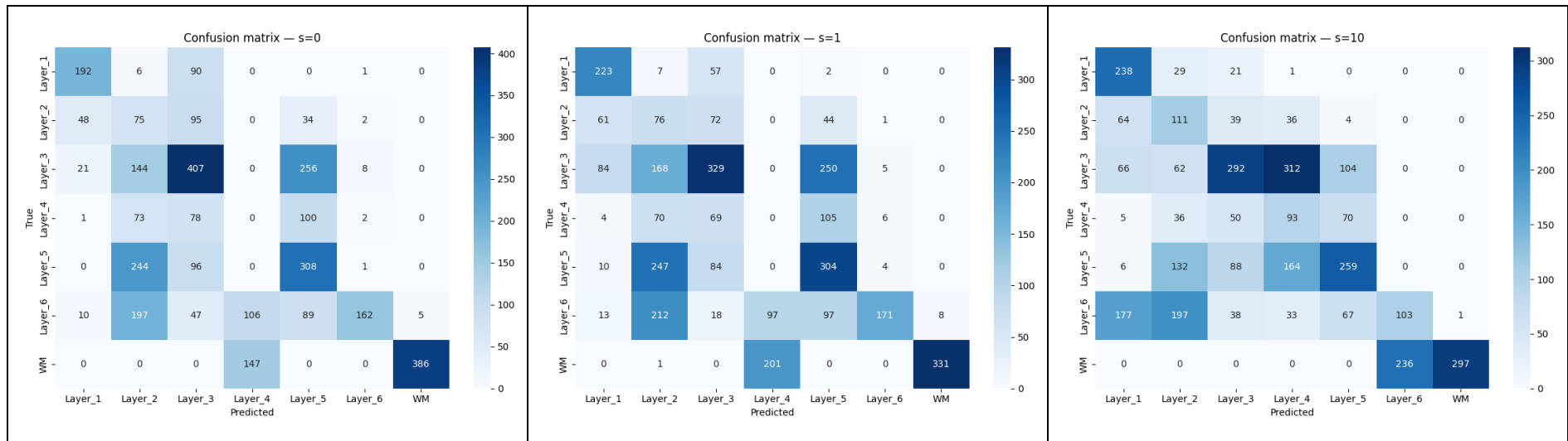


Figure 3. Confusion Matrices for SpaGCN Clustering Across Varying Image Weights.

The plots display the spot-level classification accuracy for $s = 0$, $s = 1$, and $s = 10$. Mechanistically, the purely transcriptomic approach $s = 0$ demonstrates strong performance in isolating Layer 3. Conversely, heavily weighting the histology $s = 10$ significantly improves the resolution of intermediate cortical layers; most notably, it serves as the exclusive configuration capable of detecting Layer 4, which remains entirely unresolved (F1-score of 0.00) under the other two settings.

Reference

Hu, Jian, et al. "SpaGCN: Integrating Gene Expression, Spatial Location and Histology to Identify Spatial Domains and Spatially Variable Genes by Graph Convolutional Network." *Nature Methods*, vol. 18, no. 11, 28 Oct. 2021, pp. 1342–1351, <https://doi.org/10.1038/s41592-021-01255-8>.